Aluru **Arun kumar**

Profile

MSc AI student at EPITA with experience in machine learning and deep learning using Python, TensorFlow, and scikit-learn. Published IEEE research on renewable energy forecasting and built ML pipelines using bike demand forecasting using time-series models.

Contact details

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My Portfolio
Arun949 (GitHub)
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Personal information

Citizenship: **India**

Languages: **French** (encours), **English** (B2), **Telugu** (native)

Skills

- Python, Java, SQL
- Machine Learning, Deep learning, NLP
- TensorFlow, PyTorch, Scikit-Learning,
- Pandas, NumPy, LLM, Generative AI, Apache Airflow
- Problem-Solving, Team Lead, Collaboration, Analytical Thinking

Projects

Renewable Energy Consumption on Solar and Wind Energy Prediction using Deep Learning and Generative AI

◇ Built deep learning models to predict solar and wind energy using NASA POWER dataset (2014–2024). ◇ Implemented LSTM and ARIMA models to improve prediction accuracy using feature engineering and time-series preprocessing.[GitHub]

Bike Demand Forecasting

◇ Designed a full ML pipeline with PostgreSQL-backed dataset for time-series demand prediction. ◇ Implemented SVM and Random Forest models and deployed an interactive dashboard using Streamlit for real-time predictions.[GitHub]

Automated Pneumonia Detection from Chest X-Rays Using Deep Learning

◇ Implemented transfer learning (ResNet50/EfficientNet) for multi-class X-ray classification achieving 94% accuracy and 0.97 AUC. ◇ Built a Flask web app with Grad-CAM explainability and Docker deployment for real-time predictions.[GitHub]

Grid Stability Optimisation using Reinforcement Learning

◇ Implemented a PPO-based RL agent in a custom Gymnasium environment to optimise battery usage and stabilise grid demand under renewable variability. ◇ Engineered reward functions for peak shaving and load balancing; deployed a Flask dashboard with real-time visualisation and TensorBoard monitoring.[GitHub]

Education

Master in AIS. Artificial Intelligence Systems. *EPITA.* **2025-2027**
◇ Data Science for production, Action-Learning, time-series models, natural language processing.

Bachelor of Technology. Artificial intelligence in machine learning. *Kalasalingam Academy of Research and Education.* **2021–2025**
◇ Thesis title: *Renewable Energy Consumption on Solar and Wind Energy Prediction using Deep Learning and Generative AI.*

Experience

Teachnook / Croatia Tech. *Machine Learning Intern.* **2023**
◇ Developed a Ml Model with 95% accuracy and deployed it via a Flask web app. ◇ Optimized model performance with data preprocessing and collaborated on real-world validation.

Publications

Renewable Energy Consumption on Solar and Wind Energy Prediction using Deep Learning and Generative AI - IEEE (2024).
View Publication *Automated Pneumonia Detection from Chest X-Rays Using Deep Learning - ICITISEE (2023).*